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Euclidean Geometry In Mathematical Olympiads

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Lemma 1.45 (Right Angles on Incircle Chord)



geometry

fgm

Angle Chase



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Kagebaka

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#1

The incircle of ABC is tangent to $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ at D , E , F , respectively. Let M and N be the midpoints of $\overline{BC}$ and $\overline{AC}$, respectively. Ray BI meets line EF at K . Show that $\overline{BK} \perp \overline{CK}$. Then show K lies on line MN .

Kagebaka

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Oct 4, 2018, 6:55 PM • 3 🍌

#2

Solution Part 1

Let $\angle ABC = 2x$. Since $\triangle CMN \sim \triangle CBA$ by SAS, $MN \parallel AB$, and drawing the angle bisector of $\angle ABC$, we find that $\angle BKM = x$ by alternating interior angles. However, this implies that $\angle KBC = \angle BKM = x \implies KM = BM$, so BKC must be a right triangle because the median length is half of the hypotenuse length.

I wasn't able to prove that K lies on MN , however. My approach was to prove that AFE is isosceles by proving that $\angle FKB = \frac{\angle ACB}{2}$, but so far it hasn't been working 😞

This post has been edited 2 times. Last edited by Kagebaka, Oct 4, 2018, 6:59 PM

Arthur.

133 posts

Oct 7, 2018, 5:34 PM • 2 🍌

#3

“ Kagebaka wrote:

$MN \parallel AB$, and drawing the angle bisector of $\angle ABC$, we find that $\angle BKM = x$ by alternating interior angles.

Doesn't this assume K , M and N are collinear?

Kagebaka

3001 posts

Oct 7, 2018, 8:02 PM • 2 🍌

#4

“ Arthur. wrote:

“ Kagebaka wrote:

$MN \parallel AB$, and drawing the angle bisector of $\angle ABC$, we find that $\angle BKM = x$ by alternating interior angles.

Doesn't this assume K , M and N are collinear?

No, because we're not using KM or KN or whatever at all in the solution.

Arthur.

133 posts

Oct 8, 2018, 3:08 PM • 4 🍌

#5

“ Kagebaka wrote:

“ Arthur. wrote:



Kagebaka wrote:

$MN \parallel AB$, and drawing the angle bisector of $\angle ABC$, we find that $\angle BKM = x$ by alternating interior angles.

Doesn't this assume K, M and N are collinear?

No, because we're not using KM or KN or whatever at all in the solution.

Maybe I'm missing something, but $MN \parallel AB$ does not imply $MK \parallel AB$ without this collinearity assumption, and your use of alternating interior angles is using this (you deduce $\angle BKM = x$ so yes, you are using KM in the solution ...).

Kagebaka

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#6

Oh wait darn you're right, I mistakenly thought that we were assuming that K was the